

Athanasios Kastoras

 [Thanos Kastoras](#) |  athanasios.kastoras@bsc.es

RESEARCH INTERESTS

Compilers, Computer Architecture, HPC and Embedded Systems, Accelerators.

EDUCATION

- 2019 - 2025 Diploma (Integrated Master) in ELECTRICAL AND COMPUTER ENGINEERING
University of Thessaly, Volos, Greece
Thesis: Exploiting AoS-to-SoA Memory Accesses of the RISC-V Vector Extension
GPA: **8.78/10**
- 20-26 Jul 2025 ACM EUROPE SUMMER SCHOOL ON HPC COMPUTER ARCHITECTURES FOR AI AND DEDICATED APPLICATIONS
Barcelona Supercomputing Center (BSC-CNS), Barcelona, Spain
Topics covered: HPC for AI, RISC-V, quantum computing, Parallel Systems

WORK EXPERIENCE

- March 2025-Present* | Junior Research Engineer at [Barcelona Supercomputing Center](#), Barcelona, Spain
Member of the "Compilers and Toolchains for HPC" group in the Computer Sciences Department. Extending the compiler of the vector RISC-V architecture to exploit AoS-to-SoA memory accesses of the RISC-V Vector Extension.
- Apr 2024-Sept 2024* | Undergraduate Student Researcher at [CSL](#), University of Thessaly, Greece
Roofline Analysis of High-Level Synthesis designs for Reconfigurable Platforms using the LLVM Compiler Infrastructure.
- Oct 2023-March 2024* | CPU Performance and Modeling Intern at [Arm](#), Sophia-Antipolis, France
Modelling and performance characterization of Dead Block Predictor based cache replacement policies. Extended a cycle-accurate model of an ARM processor written in C, with a generic implementation of a prediction table designed to support several dead block predictor policies.
- Oct 2022-Jan 2025* | Undergraduate Teaching Assistant at [E-CE, University of Thessaly](#), Greece
Computer Organization and Design, Fall 2022, Prof. Nikolaos Bellas. Examination of laboratory assignments on MIPs assembly programming, Verilog implementation of a MIPs CPU, and performance analysis and optimization of a large program on x86 CPUs.
Embedded Systems, Spring 2024, Prof. Nikolaos Bellas. Enabling students to implement a dynamic programming algorithm in High-Level Synthesis C and execute it on Zedboard FPGAs.
- Jun 2022-Sep 2022* | Research Intern at [CSL](#), University of Thessaly, Greece
Optimized the Raycast Hardware kernel on the FPGA. Raycast is used in real-time KinectFusion, a complex, Simultaneous Localization and Mapping (SLAM) algorithm for Robotics. Focused on getting over the memory access patterns limitations.
- Jul 2021-Aug 2021* | Research Intern at [Barcelona Supercomputing Center](#), Remote
Internship under the scope of PRACE Summer of HPC program. Worked on FTI, a fault tolerance library for distributed computing applications. Integrated lossy compression in the checkpoint/restart mechanism to reduce IO bottleneck on checkpoint/restart mechanism.

POSTERS

Athanasios Kastoras, Roger Ferrer Ibañez, Lorenzo Albano

[A stride Towards Generating Segment Accesses in RVV](#)

EuroLLVM: 2026 European LLVM Developers' Meeting, Dublin, Ireland

Athanasios Kastoras, Kevser Ildes, Kai Keller, Leonardo Bautista-Gomez

[Accelerating checkpoint/restart with lossy methods \(Poster at Student Research Symposium\)](#)

2021 IEEE 28th International Conference on High-Performance Computing, Data, and Analytics (HiPC'21), Bangalore, India.

EXTENDED ABSTRACTS

Kevser Ildes, Athanasios Kastoras, Kai Keller, Leonardo Bautista-Gomez

[Accelerating checkpoint/restart with lossy methods \(Extended Abstract\)](#)

2021 IEEE/ACM 11th Workshop on Fault Tolerance for HPC at eXtreme Scale (FTXS'21), St. Louis, USA

SKILLS

SOFTWARE	C, C++, Python, CUDA
HARDWARE	Verilog, HLS C/C++, Xilinx Tools
FRAMEWORKS	LLVM, OpenMP, Flex/Bison, MPI, Pthreads, Pytorch, Eigen
TOOLS	git, CMake, Makefile, Intel VTune, Xilinx Vitis HLS & Vivado

VOLUNTARY WORK/EXTRA CURRICULARS

- Elected Member of the Student Council of the E-CE department at the University of Thessaly (May 2024-April 2025)
- Head of Academics Committee at [14th Electrical and Computer Engineering Student Conference](#) (April 7-9 2023)
- Active Member of IEEE Student Branch Volos (2020-2022)

Teaching Assistant for Computer Organization and Design (September 2022-present)

- Examining weekly laboratory exercises (MIPS Assembly, Verilog implementation of pipelined MIPS, and more)
- Supervising midterm and final exams

Head of Academics at [14th Electrical and Computer Engineering Student Conference](#) (April 7-April 9 2023)

- Administrating the selection of the conference speakers and the communication with them
- Supervising the conference's time schedule

Member of IEEE Student Branch Volos (December 2019-present)

- Secretary of Student Branch during 2020
- Embedded Systems Engineer at HERMES team. Building an exoskeleton to participate in CYBATHLON 2024
- Organizer of the annual Linux install-fest for first-year students taking the Programming I course, October 2021-2022

CERTIFICATIONS

- Basic Life Support (BLS) Provider - European Resuscitation Council (ERC) (Valid until 15/05/2027)

LANGUAGES

GREEK:	Native Speaker
ENGLISH:	Intermediate (CEFR - B2)
SPANISH:	Conversational (B1)